

$$0 = 1 - 1 = -1 + 1 = 0$$

From Elementary School to Higher Algebras

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Question

Can you prove $0 = 0$, non-trivially?

Test your math level

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$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

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Hopf fibration is a map defined by

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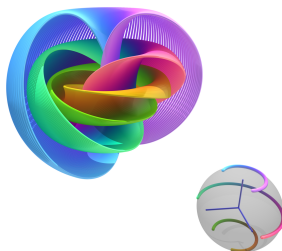
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For any point $x \in \mathbb{S}^2$, preimage $H^{-1}(x)$ is a circle $\mathbb{S}^1$. We can visualize this map by considering $H : \mathbb{R}^3 \cong \mathbb{S}^3 - \{\infty\} \rightarrow \mathbb{S}^2$:



Homotopy groups

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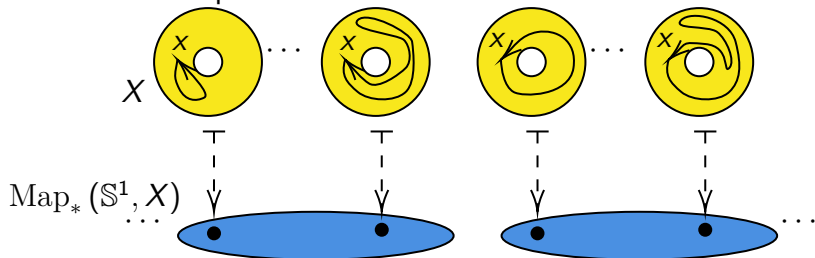
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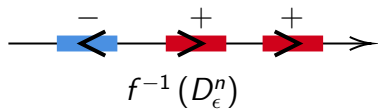
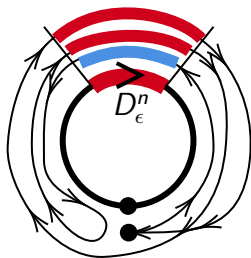
Let us study the cases $X = \mathbb{S}^n$.

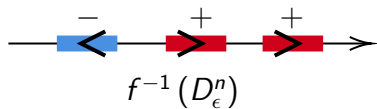
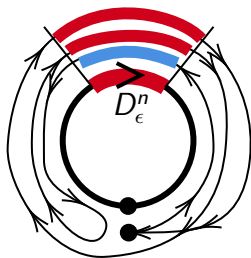
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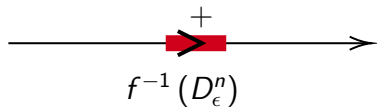
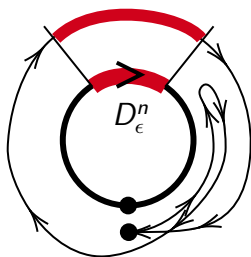
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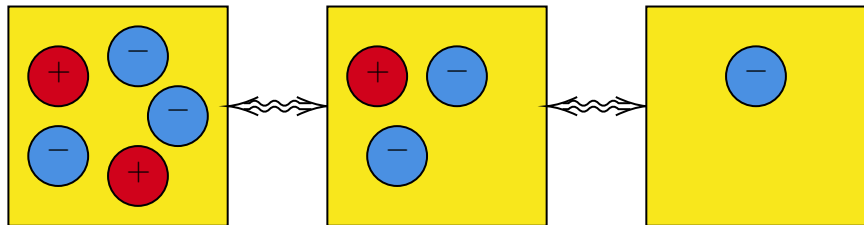
deform



Homotopy groups of spheres

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This gives an isomorphism to the configuration space of disks.

$$\text{End}_*(\mathbb{S}^n) \cong \text{Conf}^{fr}(\mathbb{R}^n) = \bigcup_{j,k} \text{Emb}(\bigsqcup_j D_+^n \sqcup \bigsqcup_k D_-^n, \mathbb{R}^n)$$

Therefore, $\pi_0 \text{End}_*(\mathbb{S}^n) \cong \mathbb{Z}$, $(j, k) \mapsto j - k$

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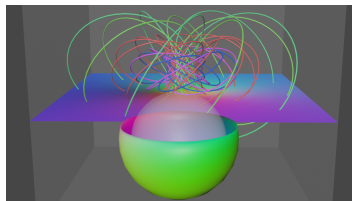
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$$\text{Map}_*(\mathbb{S}^{n+1}, \mathbb{S}^n) \cong \text{Map}_*(\mathbb{S}^1, \text{End}_*(\mathbb{S}^n)) \rightarrow \text{Map}_*([0, 1], \text{End}_*(\mathbb{S}^n))$$

Then H corresponds to a path (deformation or homotopy) from $0 \in \text{End}_*(\mathbb{S}^2)$ to 0 itself.

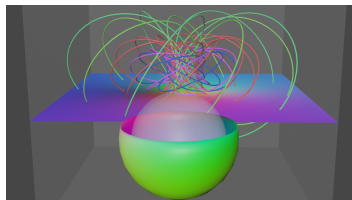
Visualize H as deformation

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We can also visualize it as a movement in the configuration space.
We will show it in the end.

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We use the **Grothendieck group** $K_0(\text{FinSet})$ (we can think finite set as $\mathbb{F}_1\text{-mod}$):

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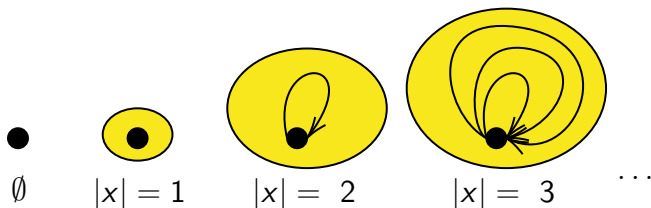
But what is $K_i(\text{FinSet})$ for $i > 0$?

Space of "finite sets"

Instead of thinking the set of "finite sets", we now consider the space of them $F = \text{FinSet}^{\cong}$: every point $x \in F$ corresponds to a finite set, and a path between two points corresponds to an isomorphism $f : x \xrightarrow{\cong} y$ of two finite sets.

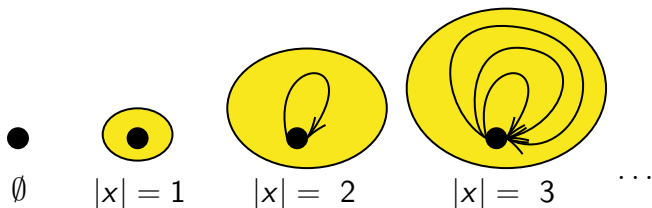
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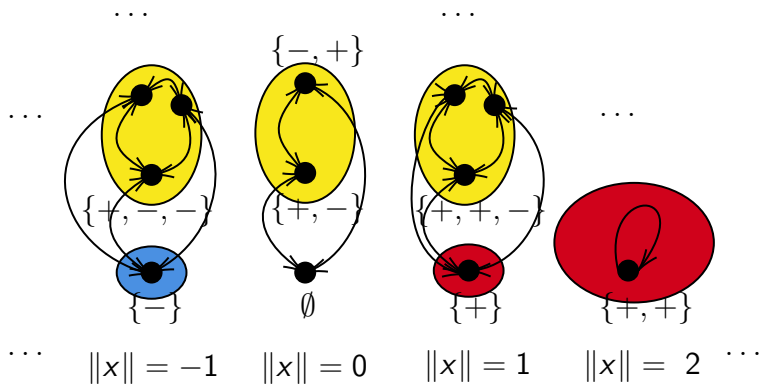
- $\pi_0(F) = \text{FinSet}_{/\cong} \cong \mathbb{N}$.
- $\pi_1(F, x) = S_{|x|}$, the permutation group.
- $\pi_i(F, x) = \{0\}$ for $i > 1$ (one can never deform an isomorphism of set to another).

K-space (spectrum) of "finite sets"

We want to add "negative" to F . We can consider the space $K(F)$ of (ordered) finite sets, whose elements are marked by $+$ or $-$, and we also want to identify the sets with same "value" $\|X\| = |X_+| - |X_-|$, and these identifications will be the paths.

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Answer

We can prove $0 = 0$ non-trivially with $h : 0 = 1 - 1 = -1 + 1 = 0!$

What's more?

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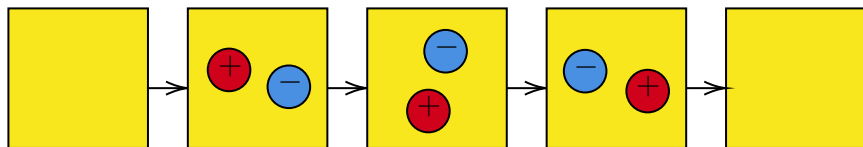
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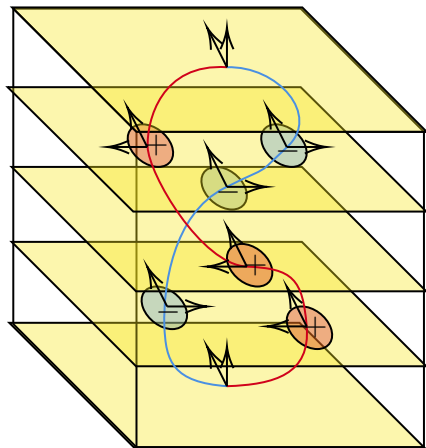
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And they are the same!



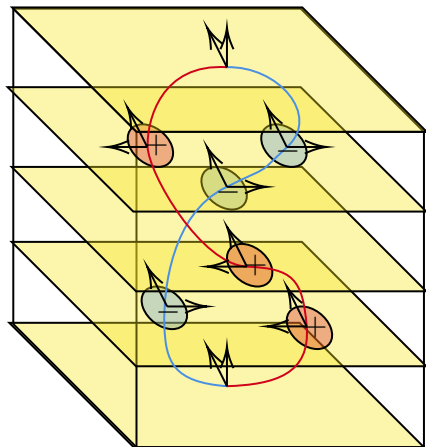
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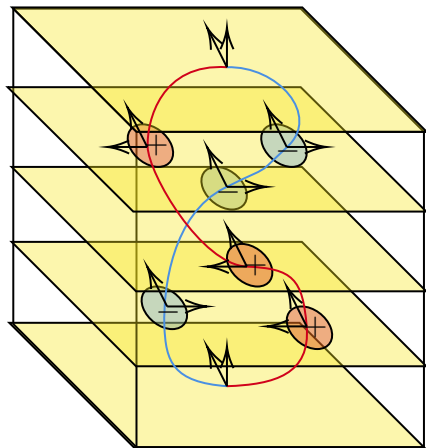
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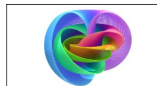
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$$\Omega_1^{fr} \cong \pi_1^s(\mathbb{S})$$

Therefore, this framed circle is non-trivial and is the generator of Ω_1^{fr} .

Thank you, hope you have got the brain upgrades!

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